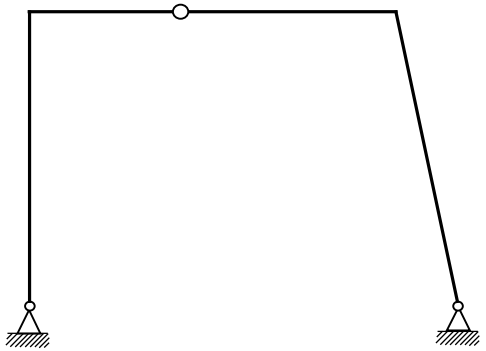
2D FRAMES

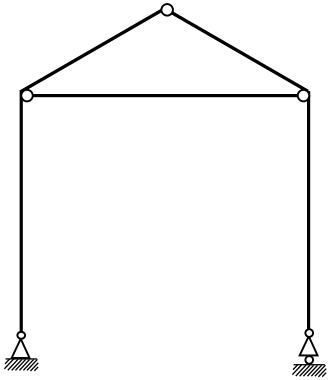
2D FRAMES



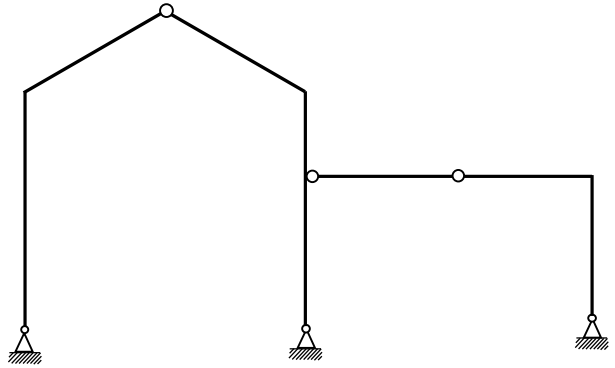
(a) Simple frame



(b) Three – hinged frame

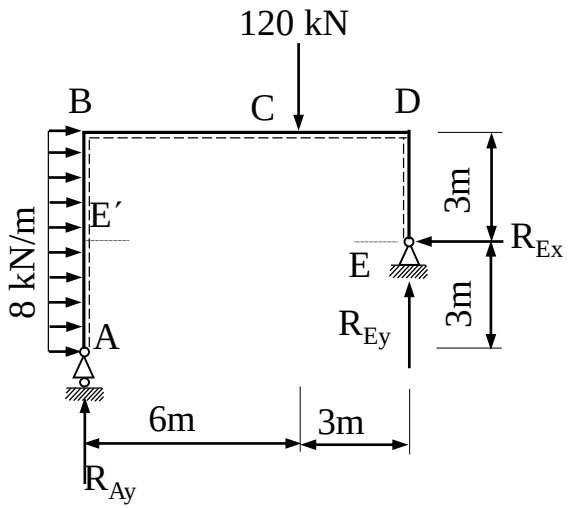


(c) Frame with tie - rod



(d) Combined three – hinged frames

Example 1:



Estimate support reactions and draw internal force diagrams of the given system.

Solution

Support Reactions

The number of unknown support reactions in horizontal direction is one

$$\sum P_x = 0 \longrightarrow -R_{Ex} + 8 \times 6{,}0 = 0 \longrightarrow R_{Ex} = 48 \ \text{kN}$$

At point E where two unknown support reactions intersects each other

$$\sum M_E = 0 \longrightarrow R_{Ay} \cdot 9{,}0 - 8 \times 6{,}0 \times 0{,}0 - 120 \times 3{,}0 = 0 \longrightarrow R_{Ay} = 40 \ \text{kN}$$

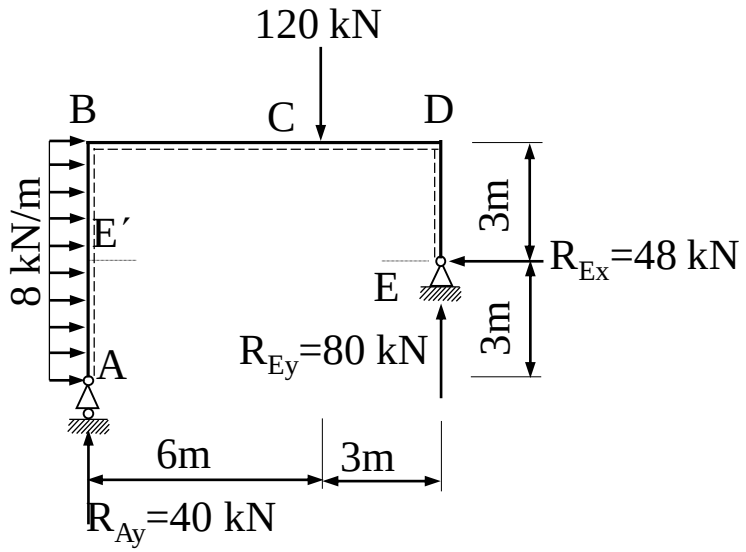
At point E′ where R_{Ay} and R_{Ex} support reactions intersects each other

$$\sum M_{E'} = 0 \longrightarrow -R_{Ey} \cdot 9{,}0 - 48{,}0 \times 0{,}0 + 120 \times 6{,}0 = 0 \longrightarrow R_{Ey} = 80 \ \text{kN}$$

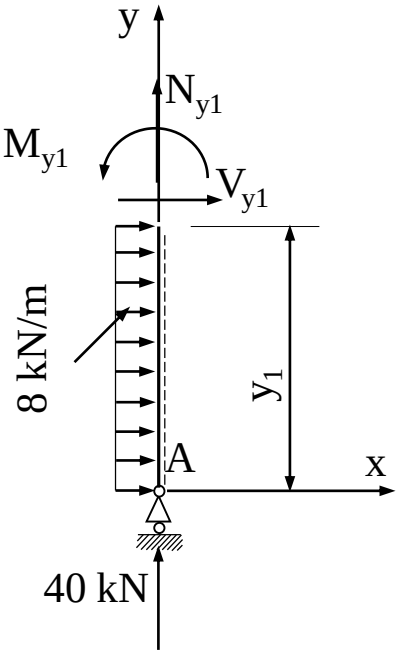
Check the results:

$$\sum P_y = 40 - 120 + 80 = 0$$

Internal Forces



AB frame



Internal forces can be written at a distance of y_1 from A as a function of y_1

$$\sum P_y = 0 \longrightarrow N_{y1} = -40 \ \text{kN}$$

$$\sum P_x = 0 \longrightarrow V_{y1} = -8y_1$$

$$M_y = 0 \longrightarrow M_{y1} = -8y_1 \cdot y_1 / 2 = -4y_1^2$$

N is constant along the frame and does not depend on y_1. M and V values can be estimated by writing the $y_1$value wherever they are wanted to be estimated

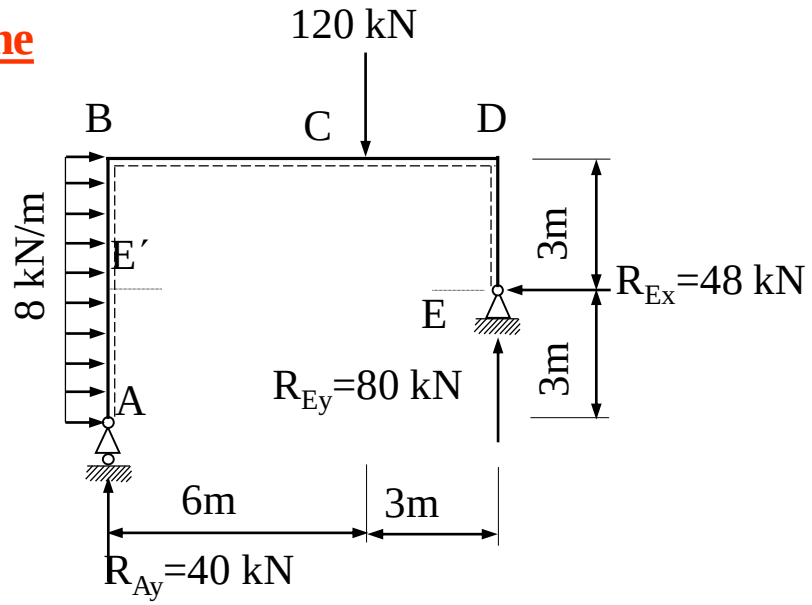
$$y_1 = 0 \longrightarrow N_{AB} = -40 \ \text{kN} \longrightarrow V_{AB} = 0$$

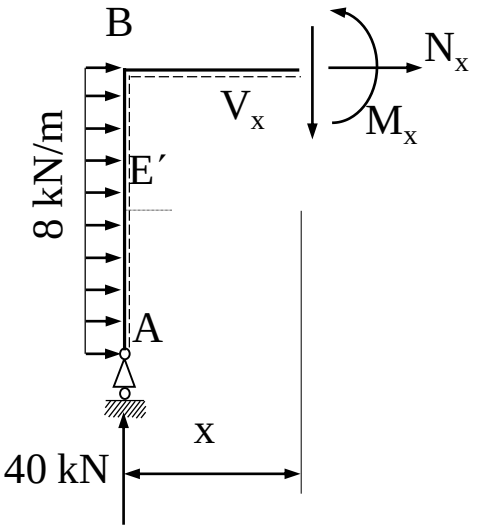
$$\longrightarrow M_{AB} = 0$$

$$y_1 = 6 \longrightarrow N_{BA} = -40 \ \text{kN} \longrightarrow V_{BA} = -48 \ \text{kN}$$

$$\longrightarrow M_{BA} = -144 \ \text{kNm}$$

BC frame





Let's cut the system between BC and write the equilibrium equations on the left system and estimate internal forces as a function of x

$$0 < x < 6\ m \longrightarrow \sum P_x = 0 \longrightarrow N_x = -8 \times 6{,}0 = -48\ kN$$

$$\sum P_y = 0 \longrightarrow V_x = 40\ kN$$

$$\sum M_x = 0 \longrightarrow M_x = 40x - 8 \times 6{,}0 \times 3{,}0 = 40x - 144$$

As it is seen from the equations, N and V are constant along the frame. Moment at point B and C can be estimated by writing the x values in the M equation

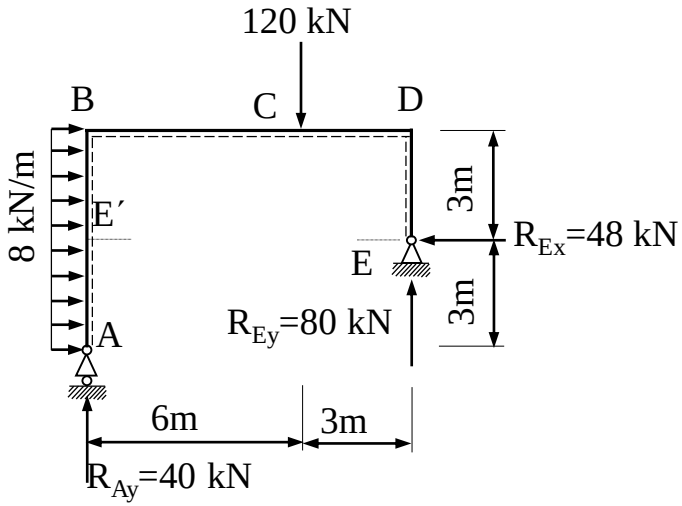
$$x = 0 \longrightarrow N_{BC} = -48\ kN \longrightarrow V_{BC} = 40\ kN$$

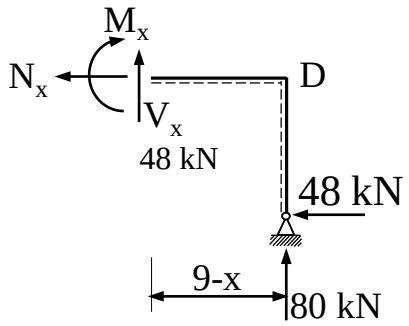
$$\longrightarrow M_{BC} = 40 \times 0{,}0 - 144 = -144\ kNm$$

$$x = 6\ m \longrightarrow N_{CB} = -48\ kN \longrightarrow V_{CB} = 40\ kN$$

$$\longrightarrow M_{CB} = 40 \times 6{,}0 - 144 = 96\ kNm$$

CD frame





Let's cut the system between CD and write the equilibrium equations on the right system and estimate internal forces as a function of x

$$6 < x < 9 \ \text{m} \longrightarrow \sum P_x = 0 \longrightarrow N_x = -48 \ \text{kN}$$

$$\longrightarrow \sum P_y = 0 \longrightarrow V_x = -80 \ \text{kN}$$

$$\longrightarrow \sum M_x = 0 \longrightarrow M_x = 80(9-x) - 48 \times 3{,}0 = 80(9-x) - 144$$

As it is seen from the equations, N and V are constant along the frame. Moment at point C and D can be estimated by writing the x values in the M equation

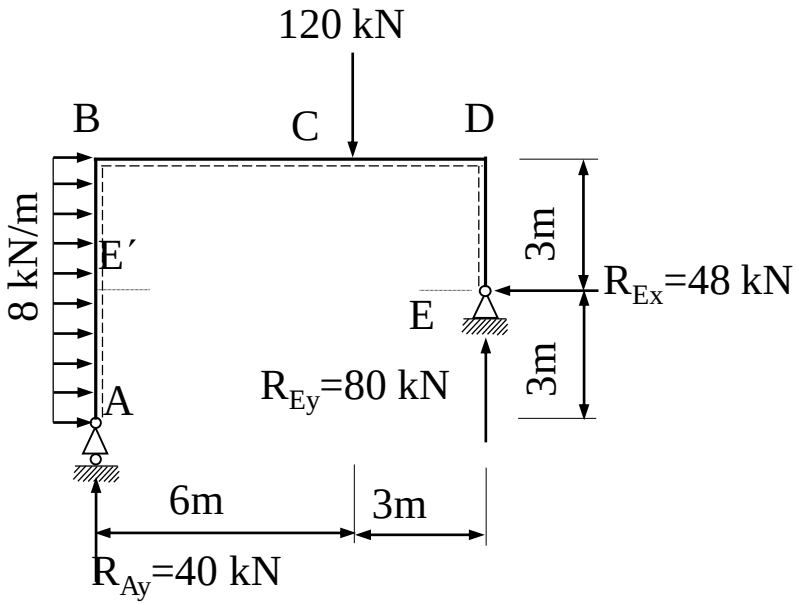
$$x = 6\,\text{m} \longrightarrow N_{CD} = -48 \ \text{kN} \longrightarrow V_{CD} = -80 \ \text{kN}$$

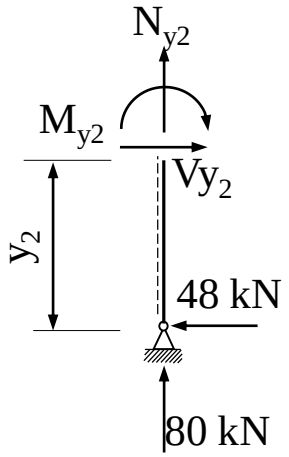
$$\longrightarrow M_{CD} = 80(9{,}0 - 6) - 144 = 96 \ \text{kNm}$$

$$x = 9 \ \text{m} \longrightarrow N_{DC} = -48 \ \text{kN} \longrightarrow V_{DC} = -80 \ \text{kN}$$

$$\longrightarrow M_{DC} = 80(9{,}0 - 9{,}0) - 144 = -144 \ \text{kNm}$$

DE frame





Let's cut the system between DE and consider the bottom system since it has lesser forces

$$0 < y_2 < 3 \ \text{m} \longrightarrow \sum P_y = 0 \longrightarrow N_{y2} = -80 \ \text{kN}$$

$$\longrightarrow \sum P_x = 0 \longrightarrow V_{y2} = 48 \ \text{kN}$$

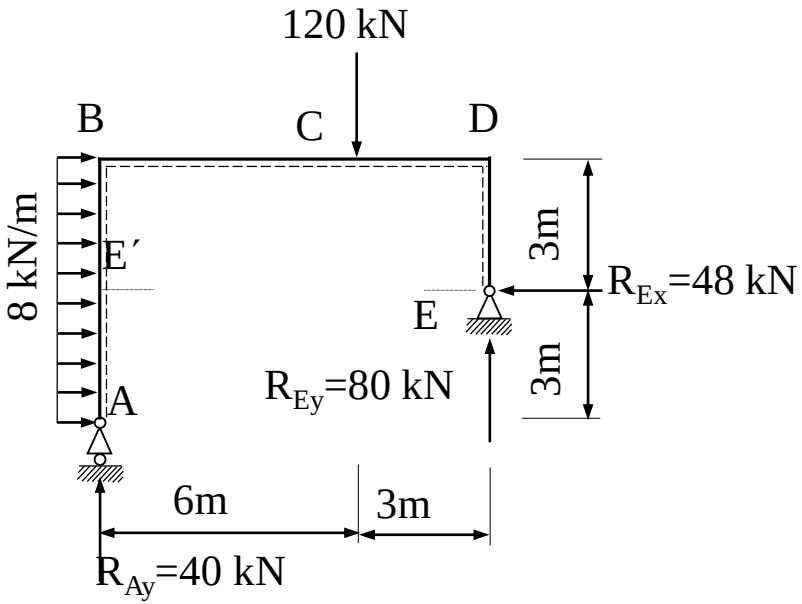
$$\longrightarrow \sum M_y = 0 \longrightarrow M_{y2} = -48y_2$$

As it is seen from the equations, N and V are constant along the frame. Moment at point D and E can be estimated by writing the y_2 values in the M equation

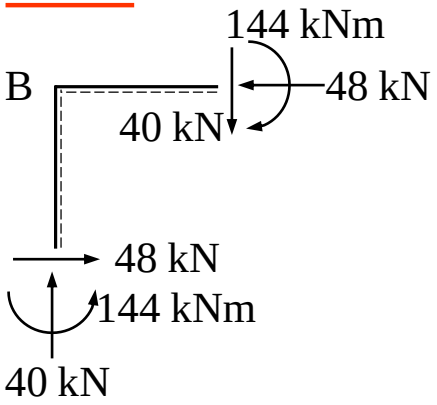
$$y_2 = 0 \longrightarrow N_{ED} = -80 \ \text{kN} \longrightarrow V_{ED} = 48 \ \text{kN} \longrightarrow M_{ED} = 0$$

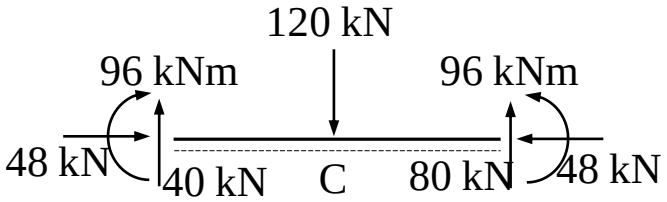
$$y_2 = 3 \ \text{m} \longrightarrow N_{DE} = -80 \ \text{kN} \longrightarrow V_{DE} = 48 \ \text{kN}$$

$$\longrightarrow M_{DE} = -48 \times 3{,}0 = -144 \ \text{kNm}$$



Check:





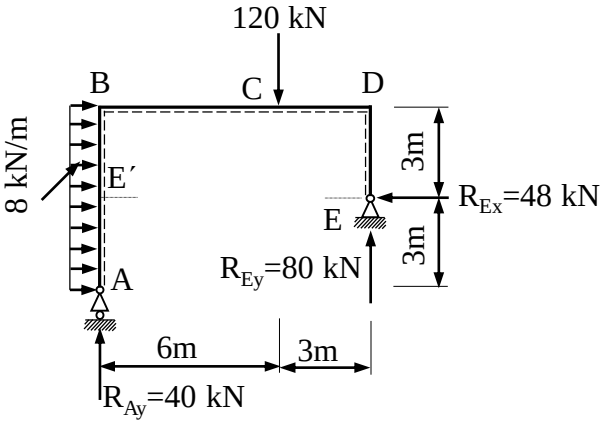
At joint B

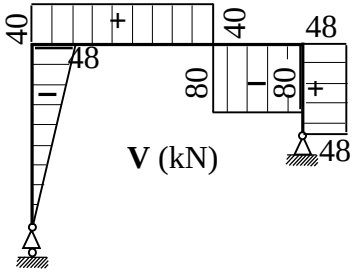
$$\sum P_x = 48 - 48 = 0$$
$$\sum P_y = 40 - 40 = 0$$
$$\sum M_B = -144 + 144 = 0$$

At joint C

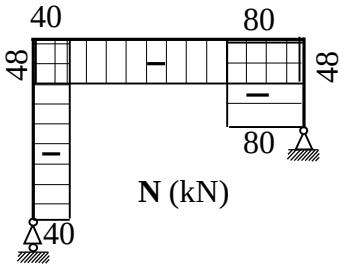
$$\sum P_x = 48 - 48 = 0$$
$$\sum P_y = 40 - 120 + 80 = 0$$
$$\sum M_B = -96 + 96 = 0$$

Estimated internal forces at each joint is correct.

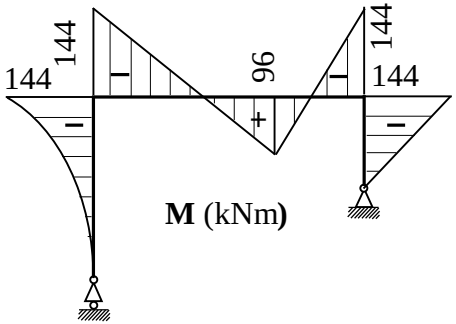




Shear Force Diagram

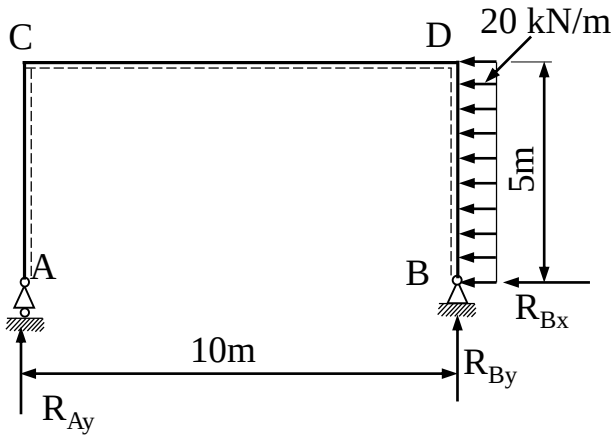


Axial Force Diagram



Moment Diagram

Example 2:



Estimate support reactions and draw internal force diagrams of the given system.

Solution:

Support Reactions

$$\sum P_x = 0 \longrightarrow R_{Bx} = -20 \times 5{,}0 = -100 \ \text{kN}$$

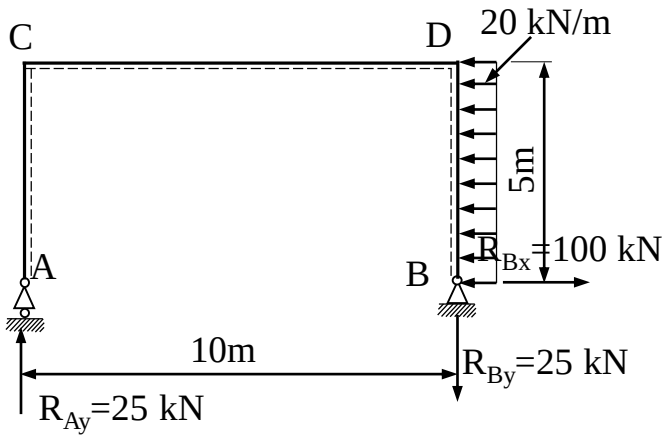
$$\sum M_B = 0 \longrightarrow R_{Ay} = (20 \times 5{,}0 \times 2{,}5)/10 = 25 \ \text{kN}$$

$$\sum M_A = 0 \longrightarrow R_{By} = (-20 \times 5{,}0 \times 2{,}5)/10 = -25 \ \text{kN}$$

Check the results:

$$\sum P_y = 25 - 25 = 0$$

Internal Forces



Since there is no external loads on AC frame:

$N_{AC} = N_{CA} = -25 \ \text{kN}$ $\qquad V_{AC} = V_{CA} = 0$

$M_{AC} = 0$ $\qquad M_{CA} = 0$

Let's cut CD frame at the right of the joint C and consider the left part:

$N_{CD} = N_{DC} = 0$ $\qquad V_{CD} = V_{DC} = 25 \ \text{kN}$ $\qquad M_{CD} = M_{CA} = 0$

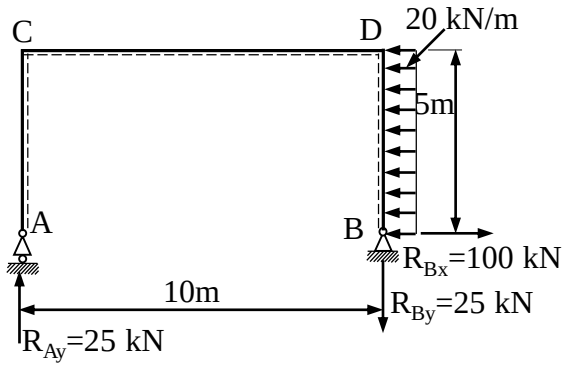
Let's cut CD frame at the left of the joint D and consider the left part:

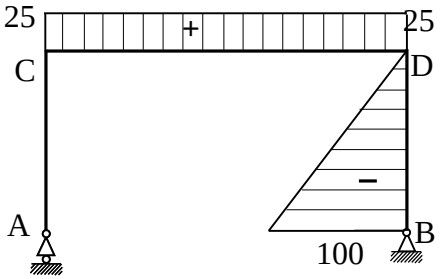
$M_{DC} = 25 \times 10{,}0 = 250 \ \text{kNm}$

Let's cut BD frame at the bottom of the joint D and consider the bottom part:

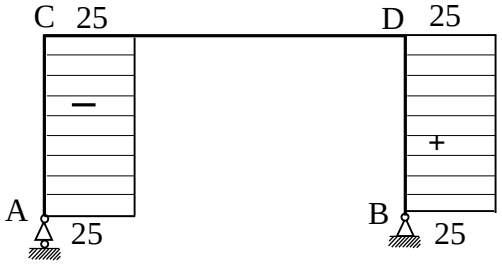
$N_{DB} = 25 \ \text{kN}$ $\qquad V_{DB} = -100 + 20 \times 5{,}0 = 0$

$M_{DB} = 100 \times 5{,}0 - 20 \times 5{,}0 \times 2{,}5 = 250 \ \text{kNm}$

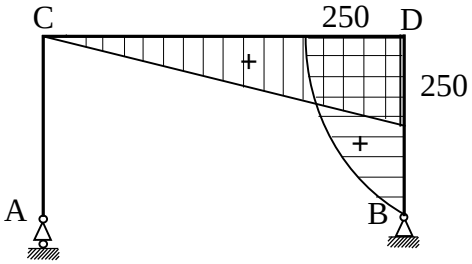




Shear force diagram (kN)

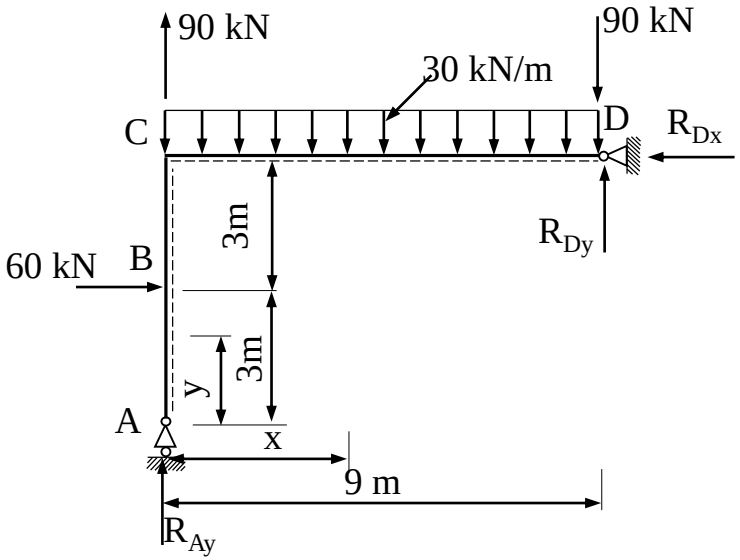


Axial force diagram (kN)



Moment diagram (kNm)

Example 3:



Estimate support reactions and draw internal force diagrams of the given system.

Solution

Support Reaction

$$\sum P_x = 0 \longrightarrow R_{Dx} = 60 \text{ kN}$$

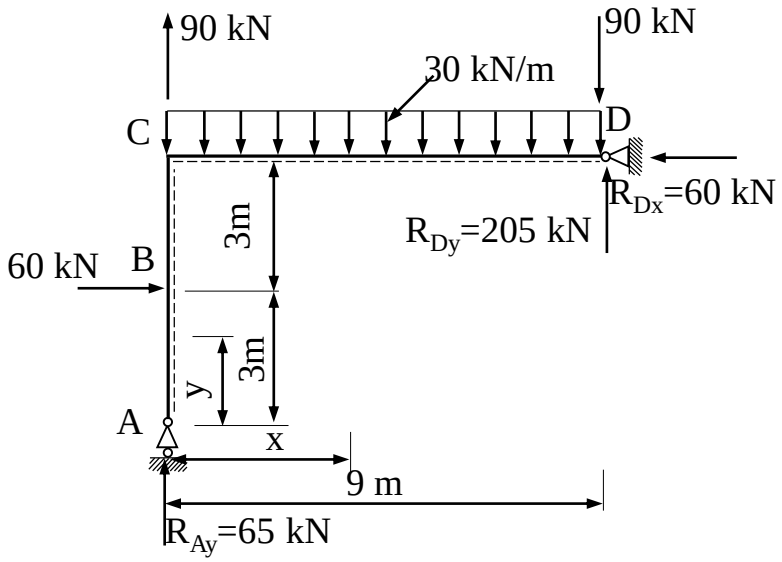
Point D where R_{Dx} and R_{Dy} support reactions intersects each other

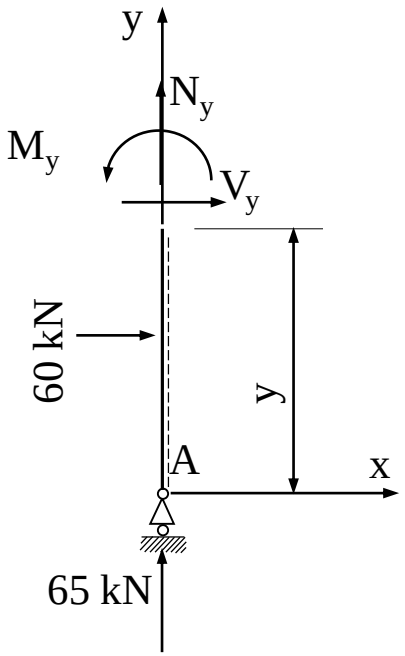
$$\sum M_D = 0 \longrightarrow R_{Ay} = (60 \times 3{,}0 - 90 \times 9{,}0 + 30 \times 9{,}0 \times 4{,}5)/9 = 65 \text{ kN}$$

Point C where R_{Dx} and R_{Ay} support reactions intersects each other

$$\sum M_C = 0 \longrightarrow R_{DY} = (-60 \times 3{,}0 + 270 \times 4{,}5 + 90 \times 9{,}0)/9 = 205 \text{ kN}$$

Internal Forces





AB frame

$0 < y < 3\,\text{m}$

$\rightarrow V_y = 0 \quad \rightarrow N_y = -65\ \text{kN}$

$\rightarrow M_y = 0$

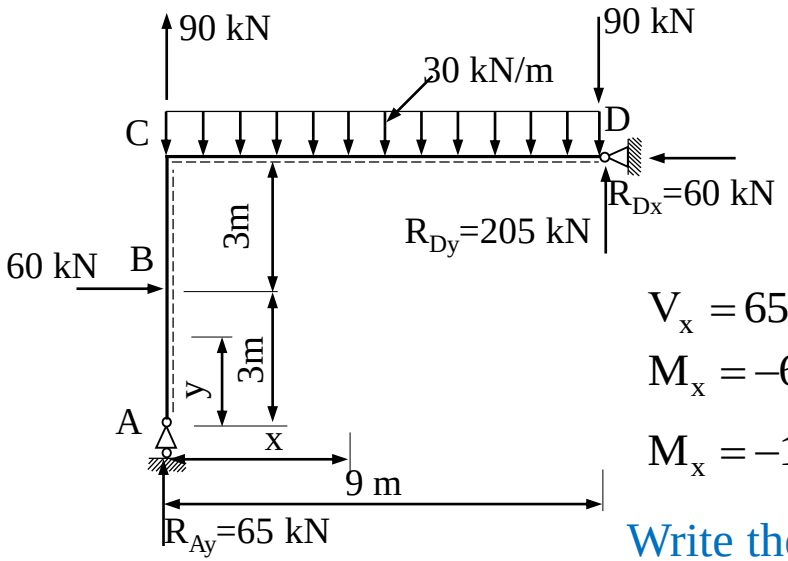
$N_{AB} = N_{BA} = -65\ \text{kN} \quad \rightarrow V_{AB} = V_{BA} = 0 \quad \rightarrow M_{AB} = M_{BA} = 0$

$3 < y < 6\,\text{m} \rightarrow V_y = -60\ \text{kN} \rightarrow N_y = -65\ \text{kN} \rightarrow M_y = -60(y-3)$

$y = 3 \rightarrow N_{BC} = -65\ \text{kN} \rightarrow V_{BC} = -60\ \text{kN} \rightarrow M_{BC} = 0$

$y = 6 \rightarrow N_{CB} = -65\ \text{kN} \rightarrow V_{CB} = -60\ \text{kN}$

$\rightarrow M_{CB} = -60(6{,}0 - 3{,}0) = -180\ \text{kNm}$



CD frame

$0 < x < 9{,}0\,\text{m}$

$N_x = -60\ \text{kN}$

$V_x = 65 + 90 - 30x = 155 - 30x$

$M_x = -60 \times 3{,}0 + (65+90)x - 30x \cdot x/2$

$M_x = -180 + 155x - 15x^2$

Write the x values into the equations

$x = 0 \longrightarrow N_{CD} = -60\ \text{kN} \longrightarrow V_{CD} = 155 - 30 \times 0{,}0 = 155\ \text{kN}$

$\longrightarrow M_{CD} = -180 + 155 \times 0 - 15 \times 0 = -180\ \text{kNm}$

$x = 9{,}0 \longrightarrow N_{DC} = -60\ \text{kN} \longrightarrow V_{DC} = 155 - 30 \times 9{,}0 = -115\ \text{kN}$

$\longrightarrow M_{CD} = -180 + 155 \times 9{,}0 - 15 \times 9{,}0^2 = 0$

Since the sign of shear force of C and D points of the frame CD are different, shear force is zero and moment is maximum at any location of span:

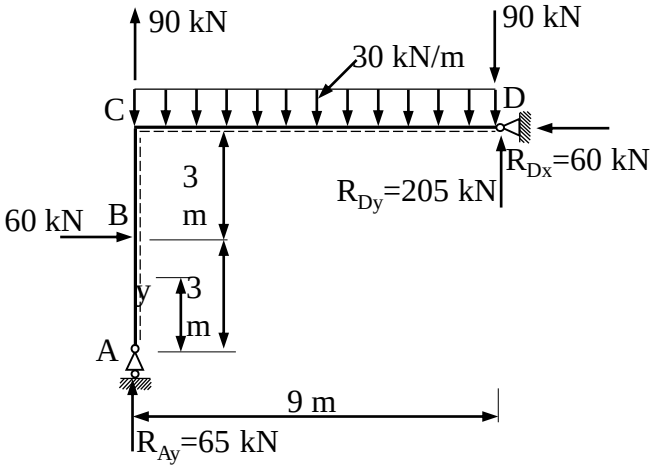
$V_x = 155 - 30x = 0 \longrightarrow x_0 = 155/30 = 5{,}167\ \text{m}$

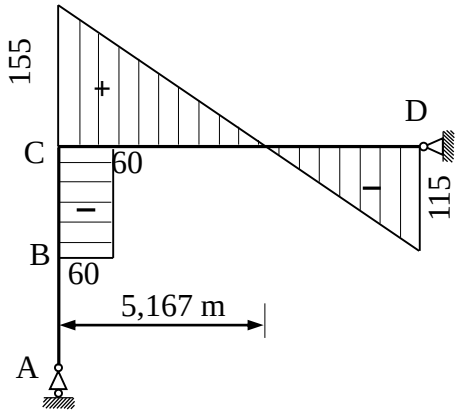
Write 5.167 m instead of x in the moment equation:

$M_{enb.} = -180 + 155 \times 5{,}167 - 15 \times 5{,}167^2 = 220{,}42\ \text{kNm}$

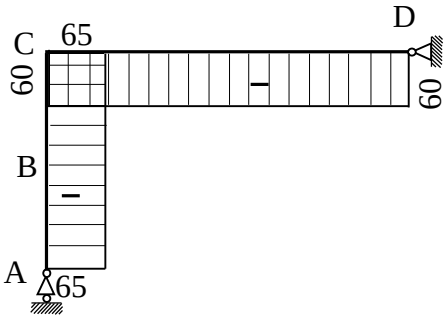
or $x_0 = 155/30 = 5{,}167\ \text{m}$

$M_{enb.} = -180 + 155^2/(2 \times 30) = 220{,}42\ \text{kNm}$

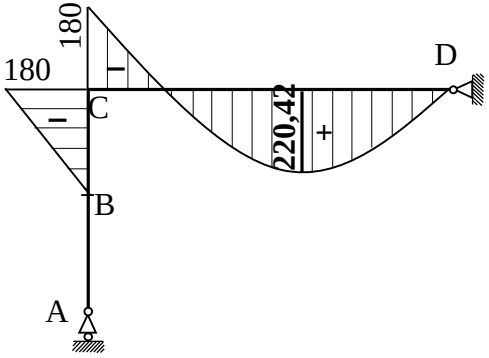




Shear force diagram (kN)

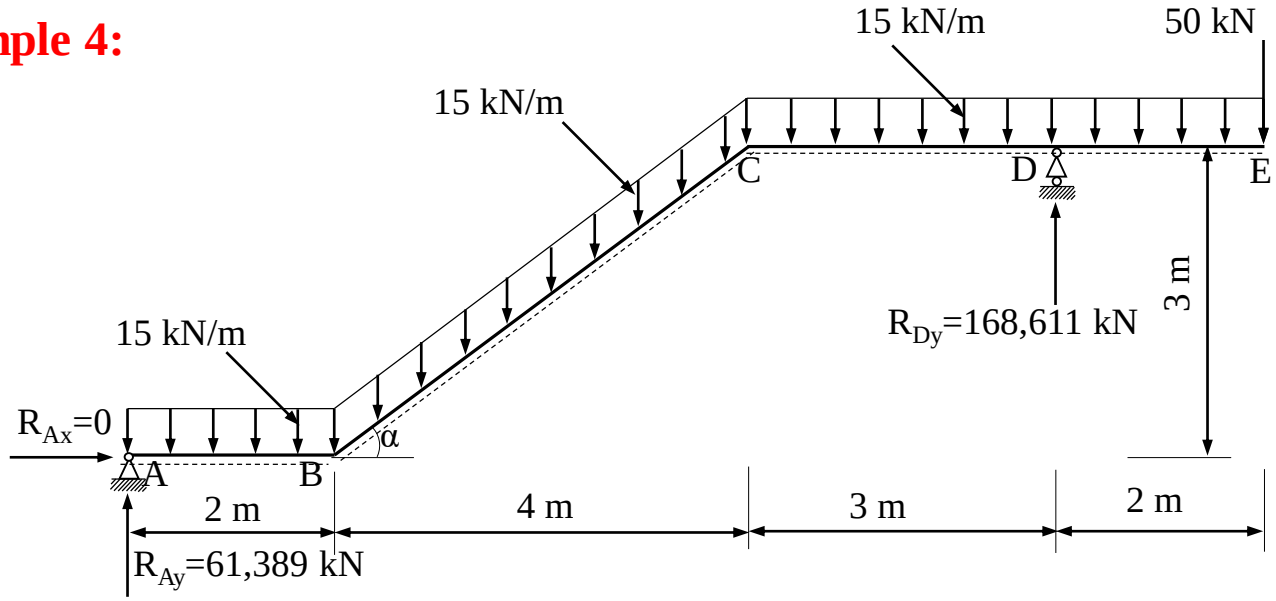


Axial force diagram (kN)



Moment diagram (kNm)

Example 4:



Estimate support reactions and draw internal force diagrams of the given system.

Solution

Length of inclined BC frame and the trigonometric identities

$L_{BC} = \sqrt{16+9} = 5 \ \text{m} \qquad \cos\alpha = 4/5 = 0{,}80 \qquad \sin\alpha = 3/5 = 0{,}60$

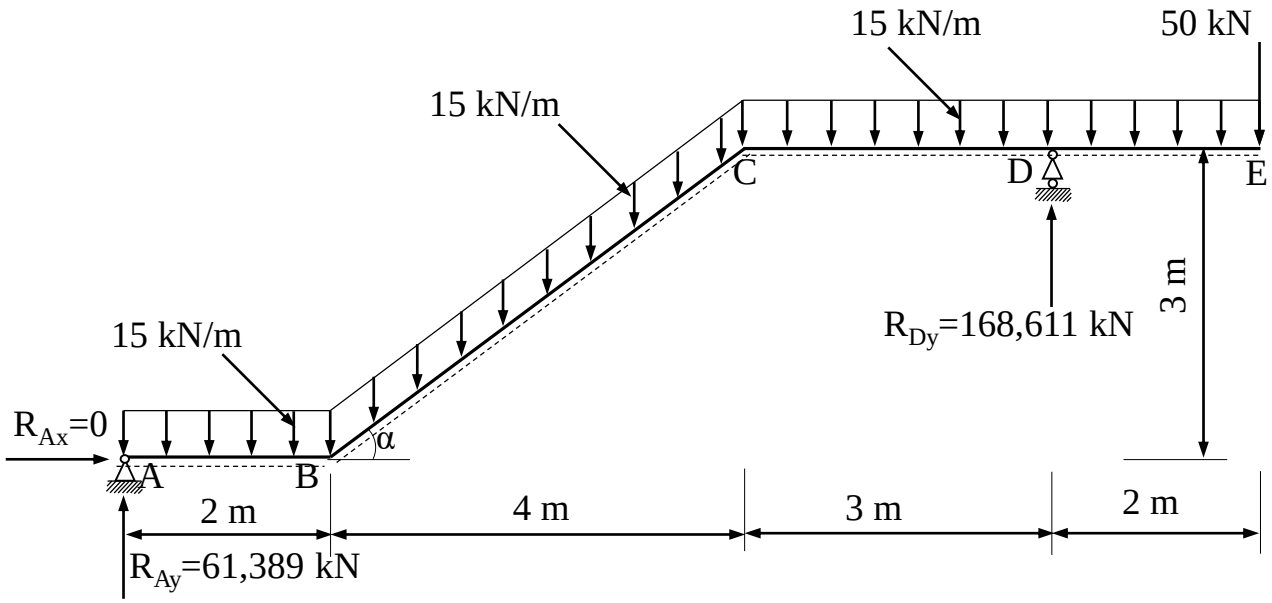
Support Reactions

$$\sum P_x = 0 \longrightarrow R_{Ax} = 0$$

$$\sum M_D = 0$$

$$R_{Ay} = (15\times 2{,}0\times 8{,}0 + 15\times 5{,}0\times 5{,}0 + 15\times 5{,}0\times 0{,}5 - 50\times 2{,}0)/9$$

$$R_{Ay} = 61{,}389 \ \text{kN}$$



$$\sum M_A = 0$$

$$R_{Dy} = (15\times 2{,}0\times 1{,}0 + 15\times 5{,}0\times 4{,}0 + 15\times 5{,}0\times 8{,}5 + 50\times 11{,}0)/9$$

$$R_{Dy} = 168{,}611 \ \text{kN}$$

Check the results:

$$\sum P_y = 61{,}389 - 30 - 75 - 75 + 168{,}611 - 50 = 0$$

Internal Forces

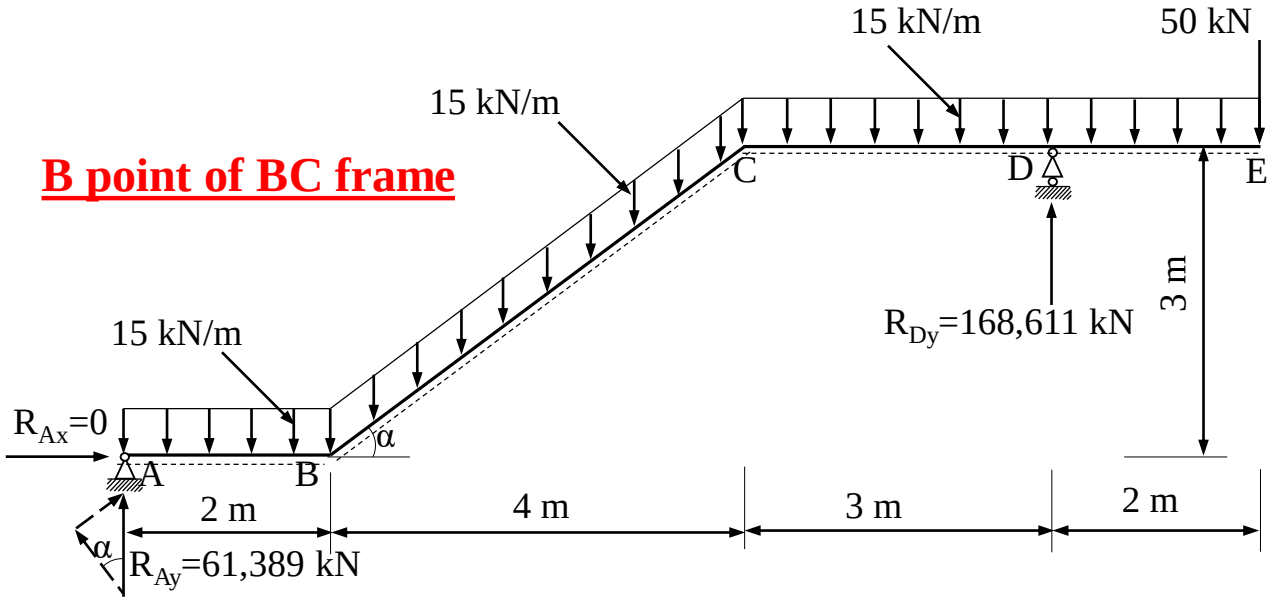
AB frame

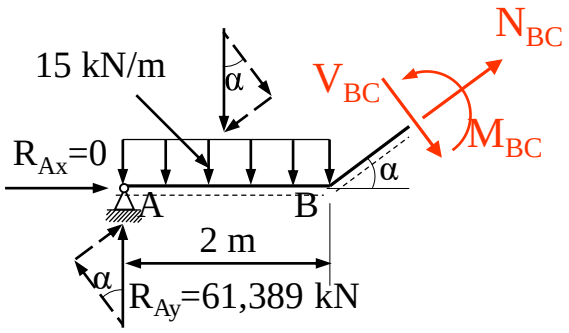
$N_{AB} = N_{BA} = 0$ $\qquad$ $V_{AB} = 61{,}389 \ \text{kN}$

$M_{AB} = 0$ $\qquad$ $V_{BA} = 61{,}389 - 30 = 31{,}389 \ \text{kN}$

$M_{BA} = 61{,}389\times 2{,}0 - 15\times 2^2/2 = 92{,}78 \ \text{kNm}$

B point of BC frame



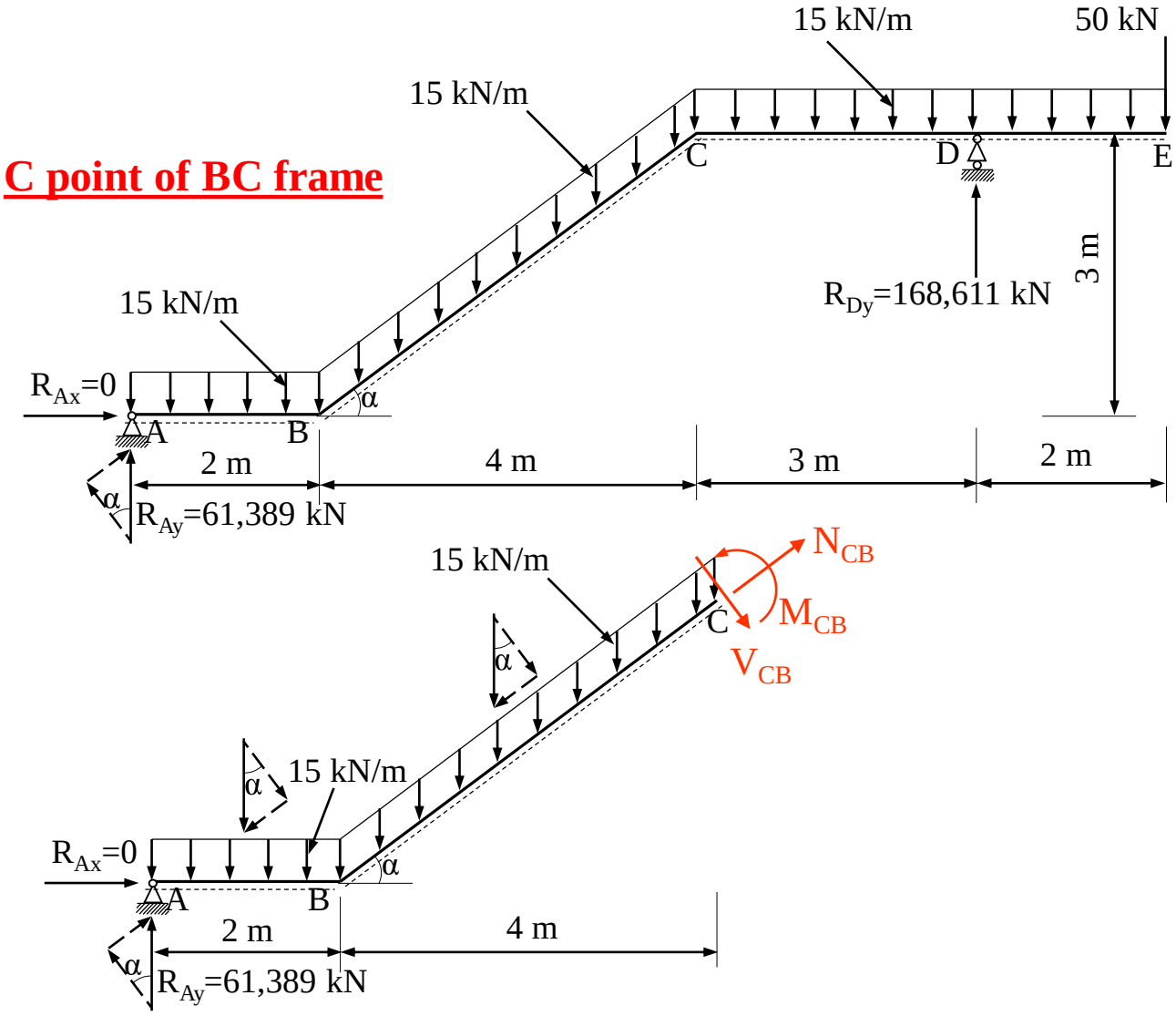


$$N_{BC} = -61{,}389 \times \sin\alpha + 15 \times 2{,}00 \times \sin\alpha = -31{,}389 \times 0{,}60 = -18{,}833 \ \text{kN}$$

$$V_{BC} = 61{,}389 \times \cos\alpha - 15 \times 2{,}00 \times \cos\alpha = 31{,}389 \times 0{,}80 = 25{,}111 \ \text{kN}$$

$$M_{BC} = M_{BA} = 92{,}78 \ \text{kNm}$$

C point of BC frame



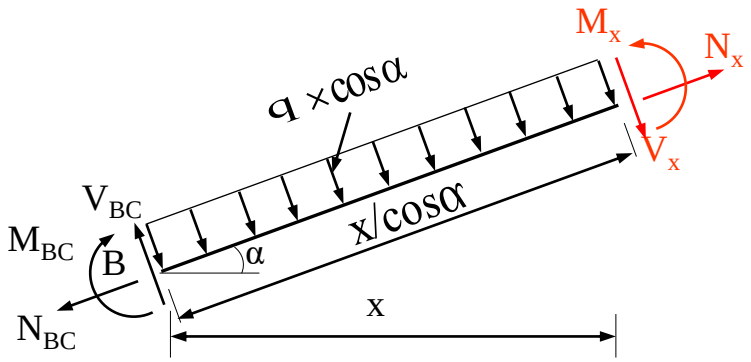
$$N_{CB} = -61{,}389 \times \sin\alpha + 15 \times 2{,}00 \times \sin\alpha + 15 \times 5{,}0 \times \sin\alpha$$

$$N_{CB} = 43{,}611 \times 0{,}60 = 26{,}167 \ \text{kN}$$

$$V_{CB} = 61{,}389 \times \cos\alpha - 15 \times 2{,}00 \times \cos\alpha - 15 \times 5{,}0 \times \cos\alpha$$

$$V_{CB} = -43{,}611 \times 0{,}8 = -34{,}889 \ \text{kN}$$

$$M_{CB} = M_{CD} = 61{,}389 \times 6{,}0 - 15 \times 2{,}0 \times 5{,}0 - 15 \times 5{,}0 \times 2{,}0 = 68{,}33 \ \text{kNm}$$



Estimation of the location of zero shear force and maximum moment

$$V_x = V_{BC} - q \times \cos\alpha \cdot (x/\cos\alpha) = 0 \longrightarrow x_0 = V_{BC}/q$$

$$M_x = M_{BC} + V_{BC} \cdot (x/\cos\alpha) - q \cdot \cos\alpha \cdot (x/\cos\alpha)(x/2\cos\alpha)$$

$$M_x = M_{BC} + V_{BC} \cdot x/\cos\alpha - q \cdot x^2/(2\cos\alpha)$$

Write the value of x instead of x_0

$$M_{enb.} = M_{BC} + V_{BC}^2/(2q \cdot \cos\alpha)$$

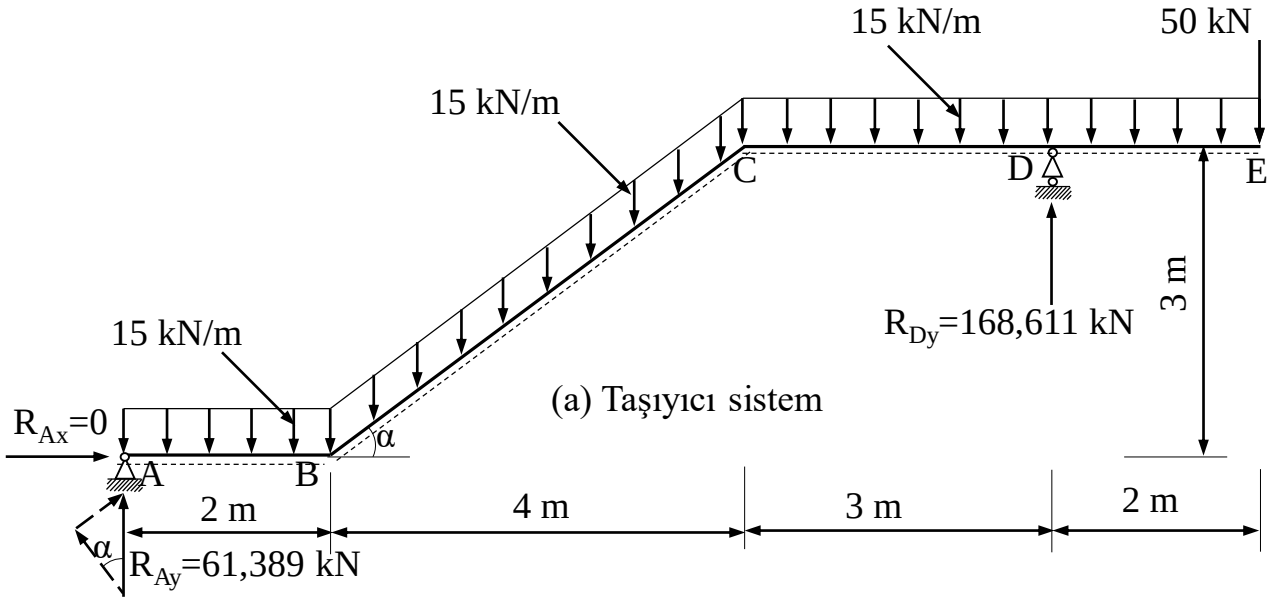
or $\quad q \cdot \cos\alpha = q_y \qquad M_{enb.} = M_{BC} + V_{BC}^2/(2q_y)$

$$x_0 = V_{BC}/15 = 25{,}111/15 = 1{,}674 \ \text{m}$$

Maximum moment

$$M_{enb.} = M_{BC} + V_{BC}^2/(2 \times 15 \cdot \cos\alpha)$$

$$M_{enb.} = 92{,}78 + 25{,}111^2/(2 \times 15 \times 0{,}8) = 119{,}05 \ \text{kNm}$$



CD frame

$$N_{CD} = N_{DC} = N_{DE} = N_{ED} = 0$$

$$V_{CD} = 15 \times 5{,}0 - 168{,}611 + 50 = -43{,}611 \ \text{kN}$$

$$V_{DC} = 15 \times 2{,}0 - 168{,}611 + 50 = -88{,}611 \ \text{kN}$$

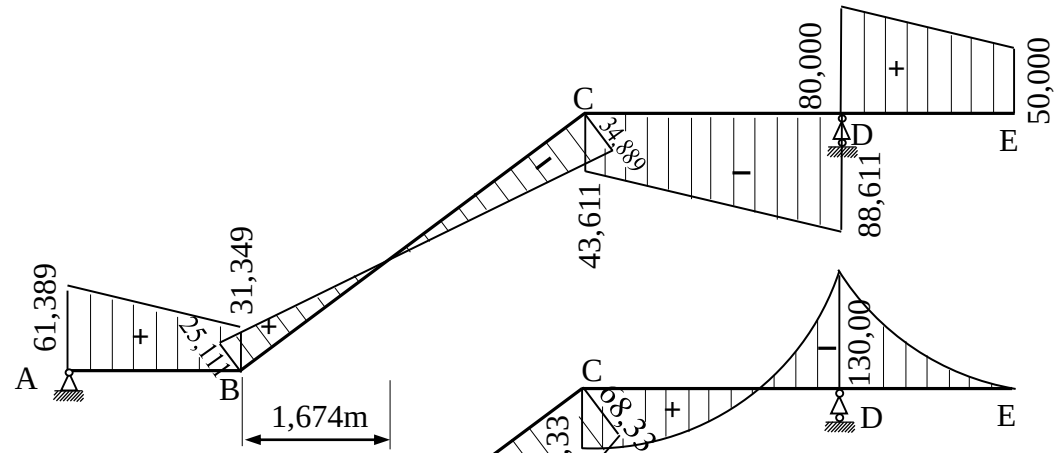
$$M_{DC} = M_{DE} = -15 \times 2{,}0^2 / 2 - 50 \times 2{,}0 = -130{,}00 \ \text{kNm}$$

DE frame

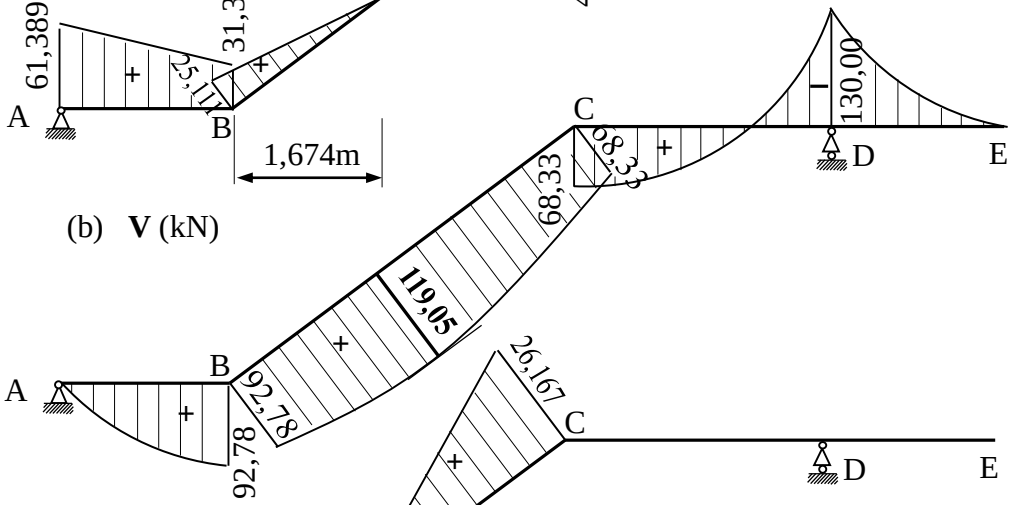
$$V_{DE} = 15 \times 2{,}0 + 50 = 80 \ \text{kN}$$

$$V_{ED} = 50 \ \text{kN}$$

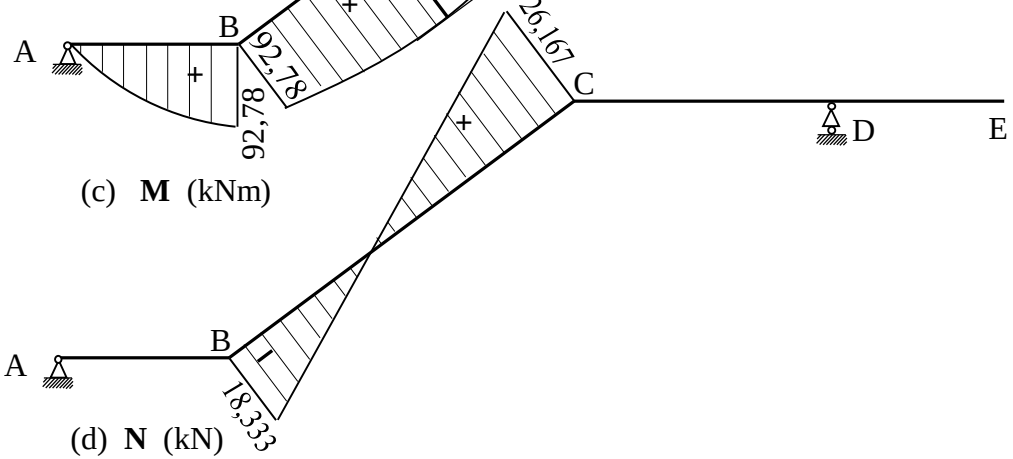
$$M_{ED} = 0$$



(b) **V** (kN)

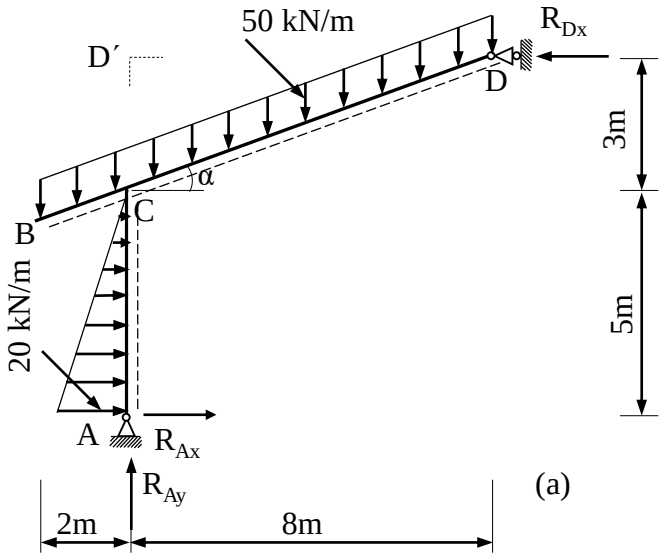


(c) **M** (kNm)



(d) **N** (kN)

Example 5:



Estimate support reactions and draw internal force diagrams of the given system.

Solution

$$L_{CD} = \sqrt{64+9} = 8{,}544 \text{ m}$$

$$\cos\alpha = 8/8{,}544 = 0{,}9363$$

$$\sin\alpha = 3/8{,}544 = 0{,}3511$$

$$L_{BC} = 2/0{,}9363 = 2{,}136 \text{ m} \qquad L_{BD} = 8{,}544 + 2{,}136 = 10{,}68 \text{ m}$$

Resultant force

$$Q_{BD} = 50 \times 10{,}68 = 534 \text{ kN} \qquad Q_{AC} = 0{,}5 \times 20 \times 5{,}0 = 50 \text{ kN}$$

Support reactions

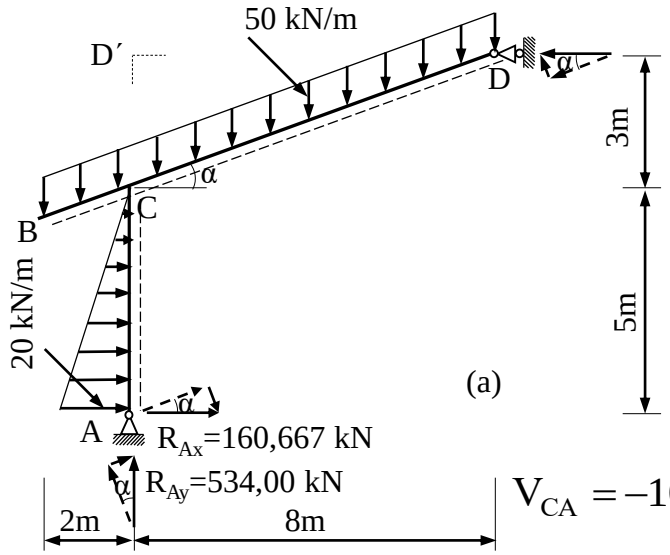
$$\sum P_y = 0 \longrightarrow R_{Ay} = 50 \times 10{,}68 = 534{,}00 \text{ kN}$$

Take moment respect to point D´ where R_{Ay} and R_{Dx} are intersected

$$\sum M_{D'} = 0 \longrightarrow R_{Ax} = [-50 \times (3{,}0 + 2 \times 5{,}0/3) + 534 \times 3{,}0]/8 = 160{,}667 \text{ kN}$$

$$R_{Ax} = 160{,}667 \text{ kN}$$

$$\sum M_A = 0 \longrightarrow R_{Dx} = [50 \times 5{,}0/3) + 534 \times 3{,}0]/8 = 210{,}667 \text{ kN}$$



Check the results:

$$\sum P_x = 160{,}667 + 50 - 210{,}667 = 0$$

Internal forces

AC frame

$$N_{AC} = N_{CA} = -534{,}00 \ \text{kN}$$

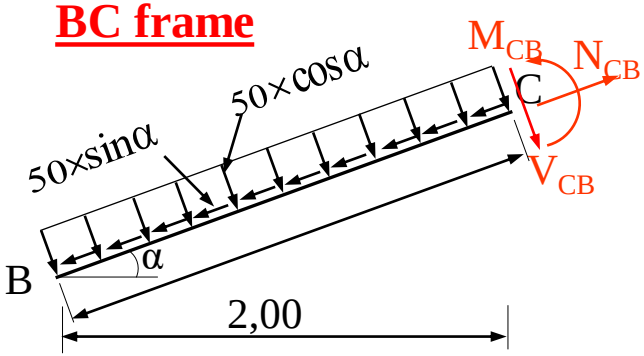
$$V_{AC} = -160{,}667 \ \text{kN}$$

$$M_{AC} = 0$$

$$V_{CA} = -160{,}667 - 0{,}5 \times 20 \times 5{,}0 = -210{,}667 \ \text{kN}$$

$$M_{CA} = -160{,}667 \times 5{,}0 - 50 \times (2 \times 5{,}0 / 3) = -970{,}00 \ \text{kNm}$$

BC frame



Resultant of distributed load parallel to the frame axis

$$q_x = 50 \cdot \sin\alpha$$

Resultant of distributed load perpendicular to the frame axis

$$q_y = 50 \cdot \cos\alpha$$

Since the point B is empty

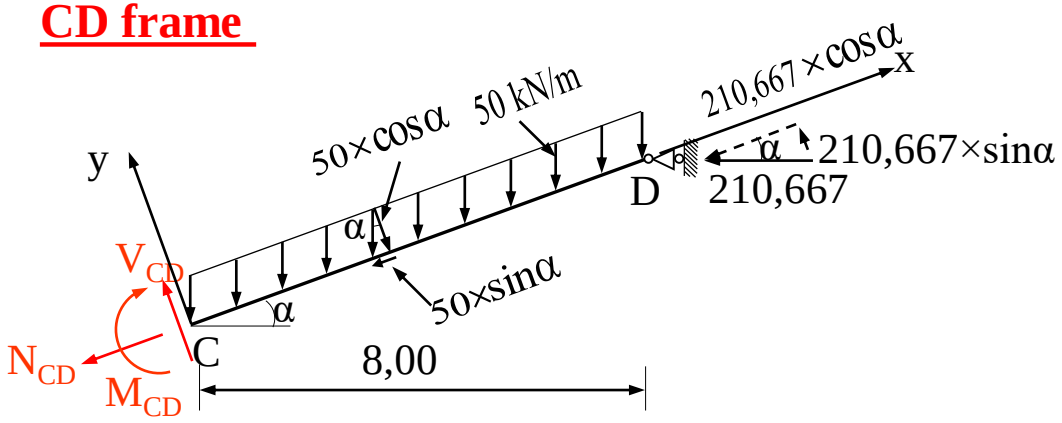
$$N_{BC} = 0 \qquad V_{BC} = 0 \qquad M_{BC} = 0$$

$$N_{CB} - 50 \times 2{,}136 \cdot \sin\alpha = 0 \qquad N_{CB} = 106{,}80 \times 0{,}3511 = 37{,}50 \ \text{kN}$$

$$V_{CB} + 50 \times 2{,}136 \cdot \cos\alpha = 0 \qquad V_{CB} = -106{,}80 \times 0{,}9363 = -100{,}00 \ \text{kN}$$

$$\sum M_C = 0 \longrightarrow M_{CB} = -50 \times 2{,}136 \times 1{,}0 = -106{,}80 \ \text{kNm}$$

CD frame



Let's cut the system from the rigth of the joint C and consider the right part:

$$N_{CD} = -50 \times 8{,}544 \cdot \sin\alpha - 210{,}667 \cdot \cos\alpha$$

$$N_{CD} = -427{,}20 \times 0{,}3511 - 210{,}667 \times 0{,}9363 = -347{,}237 \ \text{kN}$$

$$V_{CD} = 50 \times 8{,}544 \cdot \cos\alpha - 210{,}667 \cdot \sin\alpha$$

$$V_{CD} = 427{,}20 \times 0{,}9363 - 210{,}667 \times 0{,}3511 = 326{,}022 \ \text{kN}$$

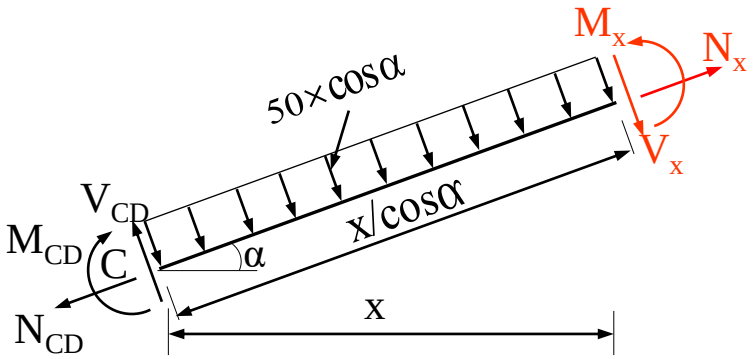
$$\sum M_C = 0$$

$$M_{CD} = -50 \times 8{,}544 \times 4{,}00 + 210{,}667 \times 3{,}0 = -1076{,}80 \ \text{kNm}$$

Let's cut the system from the left of the point D and consider the right part

$$N_{DC} = -210{,}667 \times 0{,}9363 = -197{,}248 \ \text{kN}$$

$$V_{DC} = -210{,}667 \times 0{,}3511 = -73{,}965 \ \text{kN} \qquad M_{DC} = 0$$



Since the signs of shear force of C and D points of the frame CD are different, shear force is zero and moment is maximum at any location of span:

Resultant force parallel to the frame axis does not create any moment on the system, thus only perpendicular resultant force is considered

$$V_x = V_{CD} - q \times \cos\alpha \cdot (x/\cos\alpha) = 0 \longrightarrow x_0 = V_{CD}/q$$

$$M_x = M_{CD} + V_{CD} \cdot (x/\cos\alpha) - q \cdot \cos\alpha \cdot (x/\cos\alpha)(x/2\cos\alpha)$$

$$M_x = M_{CD} + V_{CD} \cdot x/\cos\alpha - q \cdot x^2/(2\cos\alpha)$$

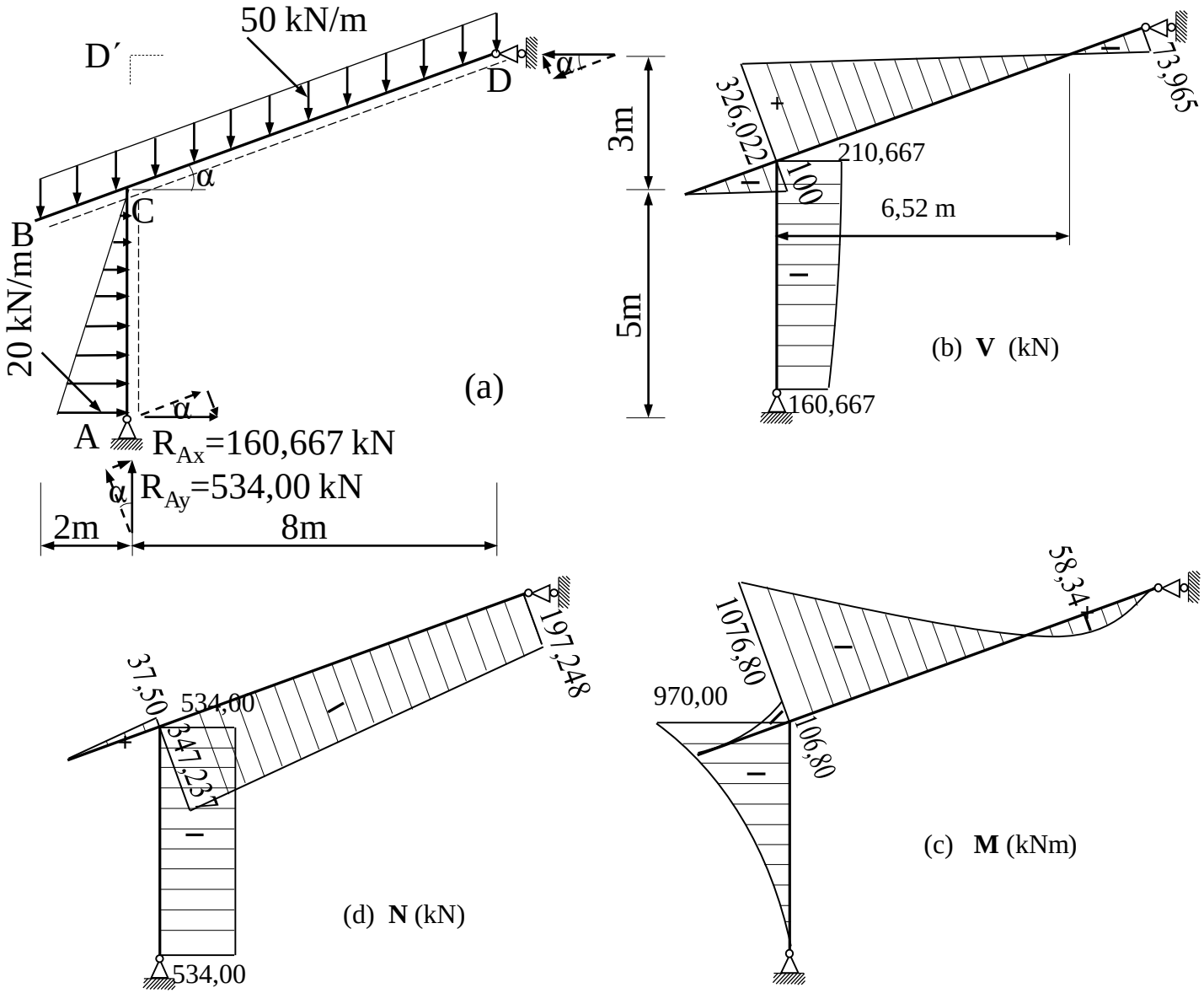
Write the value of x instead of x_0

$$M_{enb.} = M_{CD} + V_{CD}^2/(2q \cdot \cos\alpha)$$

or $q \cdot \cos\alpha = q_y$ konularak $M_{enb.} = M_{CD} + V_{CD}^2/(2q_y)$

$$x_0 = 326{,}022/50 = 6{,}52 \ \text{m}$$

$$M_{enb.} = -1076{,}80 + 326{,}022^2/(2 \times 50 \times 0{,}9363) = 58{,}43 \ \text{kNm}$$


50 kN/m
D´
D
α
C
B
20 kN/m
A
R_Ax=160,667 kN
R_Ay=534,00 kN
2m
8m
3m
5m
(a)
73,965
326,022
210,667
100
6,52 m
160,667
(b) V (kN)
58,34
1076,80
970,00
106,80
(c) M (kNm)
197,248
37,50
534,00
347,23
534,00
(d) N (kN)

Reference:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ

Acknowledgement:

Thank you to Prof. İbrahim EKİZ for sharing this presentation with me.